

Massive Pulmonary Embolism



Section I: Scenario Demographics

Scenario Title:	Massive Pulmonary Embolism		
Date of Development:	08/06/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
Affiliations/Institution(s):	McMaster University
Contact E-mail (optional):	kcaners@gmail.com

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To enhance resuscitation and team management skills using a pulmonary embolism case which requires difficult management decisions to be made quickly.
CRM Objectives:	Lead team members effectively through a challenging resuscitation.
Medical Objectives:	1) Recognize risk for PE and initiate the appropriate workup urgently. 2) Provide quality ACLS care, including: a. Using ETCO₂ to guide resuscitation b. Minimizing pulse checks c. Appropriate 30:2 compressions: breaths ratio d. Appropriate use of medications (epinephrine and thrombolytic) 3) Consider the administration of thrombolytics during cardiac arrest secondary to a suspected pulmonary embolism.

Case Summary: Brief Summary of Case Progression and Major Events

A 46 year old male with a cast on his left leg from a bad ankle fracture presents to the ED complaining of pleuritic chest pain and shortness of breath. The team will take a history and start workup when the patient will suddenly state he's "not feeling well" and then arrest. The team will perform ACLS consistent with the PEA algorithm and should consider IV thrombolytics. If IV thrombolytics are administered, the patient will have ROSC.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☒ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
		<i>Select most important dimension(s)</i>	
Confederates			
Brief Description of Role			
None.		A distraught wife could be added to the scenario to make the case more complex for senior learners.	
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☒ Capnography	
☒ Pulse Oximeter	☒ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Cast on left lower leg to simulate ankle fracture.			
E. Approximate Timing			
Set-Up:	5 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 46 year old male presents to the ED complaining of shortness of breath and pleuritic chest pain. He broke his ankle a week ago and has been in a cast since. He was just discharged home after operative repair 2 days ago.

B. Patient Profile and History

Patient Name: Vinny Kostas	Age: 46	Weight: 70kg
Gender: ☒ M ☐ F	Code Status: Full.	
Chief Complaint: SOB		
History of Presenting Illness: Twelve hours of worsening shortness of breath. Feels like it's hard to catch his breath and it gets worse when he walks. For the last two hours , he's also been having sharp, pleuritic chest pain to the right side of his chest. He's never experienced anything like this before.		
Past Medical History:	Recent ankle # with ORIF	Medications: Tylenol #3 PRN

Allergies: None.

Social History: Non-smoker.

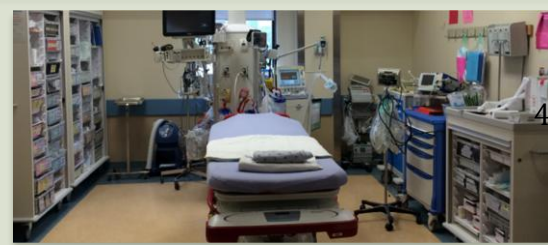
Family History: Dad has clotting disorder. Unsure what it is – makes his blood thick?

Review of Systems:	CNS:	Nil.
	HEENT:	Nil.
	CVS:	Chest pain. Pleuritic and sharp. Across his right chest. No orthopnea or PND.
	RESP:	Feel SOB. Worse when he's walking.
	GI:	Nil.
	GU:	Nil.
	MSK:	His casted leg hurts.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 115/min	BP: 95/60	RR: 18/min
O ₂ SAT: 92%		
Rhythm: Sinus tach	T: 37.2°C	Glucose: 6.3 mmol/L
GCS: 15 (E4 V5 M6)		
General Status: Looks relatively well.		
CNS:	A+O. PERLA. No FND.	
HEENT:	Nil.	
CVS:	Normal S1/S2. No murmur.	
RESP:	Good air entry bilaterally. No adventitious. Mildly tachypneic.	
ABDO:	Soft, non-tender.	
GU:	Nil.	
MSK:	Left leg in cast. No other hot/tender joints.	SKIN: Nil.

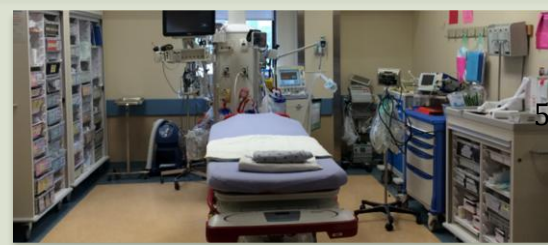
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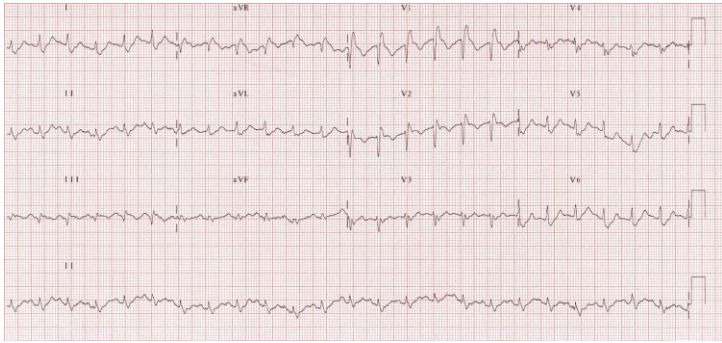
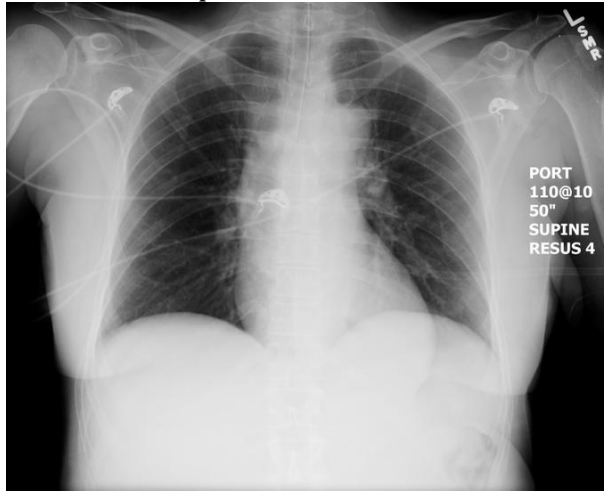
Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: HR: 115/min BP: 95/60 RR: 18/min O ₂ SAT: 92% RA T: 37.2°C	A+O. Complaining of pleuritic CP and SOB.	<u>Learner Actions</u> - ☐ IV, O ₂ , monitors - ☐ ECG - ☐ Cardiac blood work, D dimer - ☐ History and Physical - ☐ Order CT chest - ☐ ± Cardiac U/S for RV strain	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - 2 min → 2. PEA Arrest
2. PEA Arrest Rhythm → slow PEA HR → 20 BP → -/- O ₂ SAT → 68% ETCO ₂ → 10	Patient states that he doesn't feel well the becomes suddenly unconscious and pulseless. Senior Case (Optional) Wife arrives at 6 min mark and is hysterical	<u>Learner Actions</u> - ☐ Quality CPR (30:2 until intubation) - ☐ Epinephrine 1amp q3min - ☐ ± HCO ₃ and CaCl amps - ☐ Intubate - ☐ Monitor capnography - ☐ Consider thrombolytics (eg: TNK 50mg iv bolus) - ☐ Confirm dosing with local protocol/pharmacy/thrombo Senior Case (Optional) - ☐ Delegate team member to talk to wife, calm her, and explain resuscitation process	<u>Modifiers</u> - Quality CPR: ETCO ₂ → 16 - Poor CPR: ETCO ₂ → 8 <u>Triggers</u> - 3 min after thrombolytics given → 3. ROSC - 12 min → End Case
3. ROSC Rhythm → NSR HR → 85 BP → 85/45 O ₂ SAT → 96% RR → 12 (vented) ETCO ₂ → 40	Patient nonresponsive but has pulse	<u>Learner Actions</u> - ☐ ECG, check cap sugar (6.3) - ☐ Norepi or epi infusion - ☐ ± Initiate cooling - ☐ CXR - ☐ Continue thrombolytic infusion (if tPA used) - ☐ Consult ICU	<u>Modifiers</u> - Vasopressor started → BP 95/65 <u>Triggers</u> - 12 min → End Case

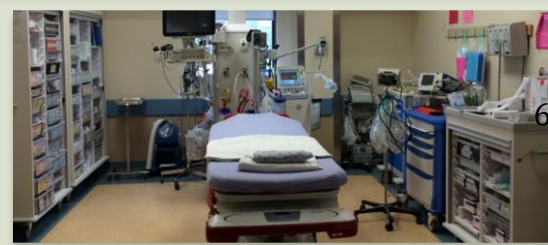
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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results	
None required for case.	
Images (ECGs, CXRs, etc.)	
ECG – Sinus tach with RBBB & TWI V1-V3 (RV strain)  http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/12/ECG-Massive-PTE.jpg	CXR – normal post-intubation male  https://emcow.files.wordpress.com/2012/11/normal-intubation2.jpg
Ultrasound Video Files (if applicable)	
U/S Showing RV strain (if done when cardiac activity)	U/S showing cardiac standstill (if done during arrest)

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) How did it feel to lead the team through this resuscitation? How did it feel to make the hard call on thrombolytics? 2) How did the team feel about the way the leader approached the thrombolytics option? 3) What are markers of quality arrest care? 4) What are the indications for thrombolytics in PE?	
Key Moments	
Recognizing possible PE.	
Coordinating arrest care.	
Decision to administer thrombolytics.	

